

Assignment 3

Part (a) - What was attempted

Note: Please take a look at the code notebook as the “graphs” are presented in an ordered manner there and are more explanatory.

1. First, we start with extracting the title and abstract from the given urls in our positive labeled dataset. We then store these strings in two distinct ways -
 - a. We concatenate our title and abstract strings using a separator token.
 - b. We keep the title and abstract embeddings separate.
2. Next, we decide on three models for generating embeddings for our title and abstract which are - BERT, SciBERT and SPECTER.
3. We save all 6 pickle dataframes for the various combinations of our models and input handling techniques.
4. Important note, we realized there are duplicates in both papers of interest and daily papers. So, they were dropped.

Techniques for generating embeddings -

1. Combine title and abstract using a separator token:
`generate_embedding(title + tokeniser.sep_token + abstract)`
2. Take mean of title and abstract embedding:
`(title_embed + abstract_embed) / 2`
3. Concatenate title and abstract embedding:
`torch.cat([title_embed, abstract_embed])`
4. We also use just title embed and just abstract_embed to use in our recommender system.

Clustering using Euclidean and Cosine distances -

1. Then we utilize the Agglomerative clustering technique to form our clusters using the positive labeled dataset. Following which, we assign clusters to our unlabelled dataset.
2. For the purpose of agglomeration, we also look at dendrograms and elbow curves to make an empirical decision of using '275' as the distance threshold.
3. Once unlabeled data is assigned to various clusters, we sort all unlabeled samples based on their distance from respective centroids and return the top 20 samples.
4. This methodology is used to get results for all the various techniques that we try out.
5. We do this for all techniques for generating embeddings and for all three models. So, in total, we have 15 unique results for each distance criteria, making it 30 in total (euclidean + cosine).

PU Techniques used:

1. To combat the problem of not having negative labels to train, we use an innovative technique in PU classification. Firstly, we initialize the positive samples to be from the labeled dataset and the entire unlabeled dataset belongs to the negative samples. Now, after each iteration, we test the unlabeled dataset and based on the confidences, we selectively choose the bottom 10 percent samples and add them to the positive dataset. This will also ensure that it is not necessary that a positive sample always remains positive and vice versa.
2. As part of the testing paradigm, we again pass the positive samples to the model and check if the model is able to recognize them as positive or negative. We do the same with the unlabeled dataset, with the expectation that most of the samples are negative and only a few are positive.
3. We do this for all techniques for generating embeddings and for all three models. So, in total, we have 15 unique results.

Part (b) - Conclusions

Clustering -

We see that for each technique, independent of the model choice, the elbow is observed at a distance of 275 out of 304. Keeping this as the threshold, we observe 30 clusters being formed for every technique. On further testing, we observe that the best papers usually belong to 3-4 unique clusters. Some common topics that keep recurring are – LLMs, In-context learning, Graphs, Reinforcement Learning, etc.

PU Classification -

We observe that the model is able to converge pretty quickly at num_iter = 10. On testing, we observe that usually, the model recognizes 90-100 papers as being positive and the remaining 800 odd to be negative. For the positive set, this value is much more skewed towards the positive side, with around 250-270 papers out of 304 being labeled positive and only 30-50 being negative. This indicates that the model is learning in the correct direction and is not biased. The model is able to learn the distribution and create a discernible distinction of both the positive and negative samples. This is supported by the fact that not all positive samples are classified as positive and not all unlabeled samples are classified as negative.